Lecture 15 (Cycling Avoidance and Lexicographic Ordering)

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If we are not careful with our pivoting rules, we may get cycling behavior. Consider the following tableau and pivoting rules

Cycling Example

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 \\ & 0 & -\frac{3}{4} & 20 & -\frac{1}{2} & 6 & 0 & 0 & 0 \\ x_5 = & 0 & \frac{1}{4} & -8 & -1 & 9 & 1 & 0 & 0 \\ x_6 = & 0 & \frac{1}{2} & -12 & -\frac{1}{2} & 3 & 0 & 1 & 0 \\ x_7 = & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Suppose we use the following pivoting rules:

1. Select the column corresponding to the most negative reduced cost to enter the basis.
2. Of all of the variables eligible to exit the basis, pick the one with the smallest index.

This will lead to the following sequence of tableaus (we will use an asterisk* to denote the pivot selected).

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 \\ & 0 & -\frac{3}{4} & 20 & -\frac{1}{2} & 6 & 0 & 0 & 0 \\ x_5 = & 0 & \frac{1}{4}^* & -8 & -1 & 9 & 1 & 0 & 0 \\ x_6 = & 0 & \frac{1}{2} & -12 & -\frac{1}{2} & 3 & 0 & 1 & 0 \\ x_7 = & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 \\ & 0 & 0 & -4 & -\frac{7}{2} & 33 & 3 & 0 & 0 \\ x_1 = & 0 & 1 & -32 & -4 & 36 & 4 & 0 & 0 \\ x_6 = & 0 & 0 & 4^* & \frac{3}{2} & -15 & -2 & 1 & 0 \\ x_7 = & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 \\ & 0 & 0 & 0 & -2 & 18 & 1 & 1 & 0 \\ x_1 = & 0 & 1 & 0 & 8^* & -84 & -12 & 8 & 0 \\ x_2 = & 0 & 0 & 1 & \frac{3}{8} & -\frac{15}{4} & -\frac{1}{2} & \frac{1}{4} & 0 \\ x_7 = & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 \\ & 0 & \frac{1}{4} & 0 & 0 & -3 & -2 & 3 & 0 \\ x_3 = & 0 & \frac{1}{8} & 0 & 1 & -\frac{21}{2} & -\frac{3}{2} & 1 & 0 \\ x_2 = & 0 & -\frac{3}{64} & 1 & 0 & \frac{3}{16}^* & \frac{1}{16} & -\frac{1}{8} & 0 \\ x_7 = & 1 & -\frac{1}{8} & 0 & 0 & \frac{21}{2} & \frac{3}{2} & -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 \\ & 0 & -\frac{1}{2} & 16 & 0 & 0 & -1 & 1 & 0 \\ x_3 = & 0 & -\frac{5}{2} & 56 & 1 & 0 & 2^* & -6 & 0 \\ x_4 = & 0 & -\frac{1}{4} & \frac{16}{3} & 0 & 1 & \frac{1}{3} & -\frac{2}{3} & 0 \\ x_7 = & 1 & \frac{5}{2} & -56 & 0 & 0 & -2 & 6 & 1 \end{bmatrix}$$

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 \\ & 0 & -\frac{7}{4} & 44 & \frac{1}{2} & 0 & 0 & -2 & 0 \\ x_5 = & 0 & -\frac{5}{4} & 28 & \frac{1}{2} & 0 & 1 & -3 & 0 \\ x_4 = & 0 & \frac{1}{6} & -4 & -\frac{1}{6} & 1 & 0 & \frac{1}{3}^* & 0 \\ x_7 = & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 \\ & 0 & -\frac{3}{4} & 20 & -\frac{1}{2} & 6 & 0 & 0 & 0 \\ x_5 = & 0 & \frac{1}{4} & -8 & -1 & 9 & 1 & 0 & 0 \\ x_6 = & 0 & \frac{1}{2} & -12 & -\frac{1}{2} & 3 & 0 & 1 & 0 \\ x_7 = & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Notice we get back the original tableau (meaning we have cycling behavior!). This example may seem a little convoluted, but in practice cycling occasionally happens and if it isn't prevented, this behavior will waste a lot of computer time. Ideally, we would like a set of rules that does not result in cycling and we would like to prove that this set of rules prevents cycling. There are two common methods for cycling avoidance.

1. Lexicographic Pivoting (We usually use this with the tableau implementation of the simplex method)
2. Bland's Rule (We usually use this with the refined simplex method)

We will prove some results about the lexicographic ordering and use this to prove that lexicographic pivoting will yield a cycle free algorithm. We will also state Bland's rule without a proof of no cycling (This proof is a little more involved than I care to cover in class. It is easily google-able, and can be understood with some linear algebra for those who are curious).

Lexicographic (Dictionary) Order

Consider alphabetizing (as we see in the dictionary). The word, "Abacus" comes before "Abdicate". Both words come before "telephone". The idea is to compare letter by letter until the first letter differs between the two words. This differing letter is the letter that dictates the order. We can do something similar for vectors. Consider the vectors,
$(1,1,1,1), (1,1,3,4), (2,1,0,0)$. If we use $\overset{L}{<}$ to denote the **Lexicographic Order**, then

$$(1,1,1,1) \overset{L}{<} (1,1,3,4) \overset{L}{<} (2,1,0,0)$$

In the first inequality, $(1,1,1,1) \overset{L}{<} (1,1,3,4)$, we see that the first two entries are the same, but in the third entry, $1 < 3$, so we see $(1,1,1,1) \overset{L}{<} (1,1,3,4)$. For the second inequality, $(1,1,3,4) \overset{L}{<} (2,1,0,0)$, we only need to look at the first entry as $1 < 2$. An equivalent (and often nicer) definition of the lexicographic order for vectors states:

$$\vec{x} \overset{L}{<} \vec{y}$$

if the first non-zero entry of $\vec{y} - \vec{x}$ is positive. If $\vec{x} \overset{L}{>} \vec{0}$, then we say that $\vec{x}$ is **Lexicographically Positive**. If $\vec{x} \overset{L}{<} \vec{y}$ or $\vec{x} = \vec{y}$, then we denote this as $\vec{x} \overset{L}{\leq} \vec{y}$.

Some Properties of the Lexicographic Order

Total Order

The lexicographic order is a total order. This means it obeys the following rules:

1. $\vec{x} \overset{L}{\leq} \vec{x}\ \forall \vec{x} \in \mathbb{R}^n$ (reflexive)
2. If $\vec{x} \overset{L}{\leq} \vec{y}$ and $\vec{y} \overset{L}{\leq} \vec{z}$ then $\vec{x} \overset{L}{\leq} \vec{z}$ (transitive)
3. If $\vec{x} \overset{L}{\leq} \vec{y}$ and $\vec{x} \overset{L}{\geq} \vec{y}$ then $\vec{x} = \vec{y}$ (antisymmetric)
4. For all $\vec{x}, \vec{y} \in \mathbb{R}^n$ either $\vec{x} \overset{L}{\leq} \vec{y}$ or $\vec{x} \overset{L}{\geq} \vec{y}$ (totality)

Let's prove these claims!

Proof

1) Suppose $\vec{x} \in \mathbb{R}^n$. Then $\vec{x} = \vec{x}$, so $\vec{x} \overset{L}{\leq} \vec{x}$.

2) Suppose $\vec{x} \overset{L}{\leq} \vec{y}$ and $\vec{y} \overset{L}{\leq} \vec{z}$. Let i be the smallest index such that $x_i < y_i$. Similarly, let j be the smallest index such that $y_j < z_j$. Then we have three cases to consider. First, suppose $i = j$. Then

$$x_i < y_i < z_i$$

and for all $k < i$ we see

$$x_k = y_k = z_k$$

so $\vec{x} \overset{L}{<} \vec{z}$. Now, suppose $i < j$. Then

$$x_i < y_i = z_i$$

and we have

$$x_k = y_k = z_k \ \forall k < i$$

so $\vec{x} \overset{L}{<} \vec{z}$. Now, suppose $i > j$. Then

$$x_j = y_j < z_j$$

and we have

$$x_k = y_k = z_k \ \forall k < j$$

so $\vec{x} \overset{L}{<} \vec{z}$.

3) Suppose $\vec{x} \overset{L}{\leq} \vec{y}$ and $\vec{x} \overset{L}{\geq} \vec{y}$ but $\vec{x} \neq \vec{y}$. Then $\vec{x} \overset{L}{<} \vec{y}$ and $\vec{x} \overset{L}{>} \vec{y}$, which implies the first non-zero entry of $\vec{x} - \vec{y}$ is positive and the first non-zero entry of $\vec{y} - \vec{x} = -(\vec{x} - \vec{y})$ is positive. This is a contradiction, so we must have $\vec{x} = \vec{y}$ if $\vec{x} \overset{L}{\leq} \vec{y}$ and $\vec{x} \overset{L}{\geq} \vec{y}$.

4) If $\vec{x} = \vec{y}$, then $\vec{x} \overset{L}{\leq} \vec{y}$ and $\vec{x} \overset{L}{\geq} \vec{y}$. Suppose now that $\vec{x} \neq \vec{y}$. Then there exists i such that $x_i \neq y_i$. Let i be the smallest such index. Then if $x_i < y_i$, we have $\vec{x} \overset{L}{\leq} \vec{y}$. If instead, we have $x_i > y_i$, we have $\vec{x} \overset{L}{\geq} \vec{y}$.

Combining Inequalities

Just like the real numbers, we have some convenient properties for combining inequalities involving the lexicographic ordering.

1. If $\vec{x} \overset{L}{\leq} \vec{y}$ then for any $\vec{z} \in \mathbb{R}^n$, then $\vec{x} + \vec{z} \overset{L}{\leq} \vec{y} + \vec{z}$.
2. If $\lambda \in \mathbb{R}$ such that $\lambda > 0$ and $\vec{x} \overset{L}{<} \vec{y}$, then $\lambda\vec{x} \overset{L}{<} \lambda\vec{y}$.
3. If $\lambda \in \mathbb{R}$ such that $\lambda < 0$ and $\vec{x} \overset{L}{<} \vec{y}$, then $\lambda\vec{x} \overset{L}{>} \lambda\vec{y}$.

Proof

The proof of **1)** and **2)** will be left as homework problems. We will prove **3)** here. Suppose $\lambda \in \mathbb{R}$ such that $\lambda < 0$ and $\vec{x} \overset{L}{<} \vec{y}$. Then there exists i such that $x_i < y_i$. Suppose i is the smallest such i. Then

$$\lambda x_i > \lambda y_i$$

and for all $k < i$, we have

$$x_k = y_k$$

thus

$$\lambda x_k = \lambda y_k$$

so

$$\lambda\vec{x} \overset{L}{>} \lambda\vec{y}$$

Lexicographic Pivoting Rule

The **Lexicographic Pivoting Rule** is stated below:

1. Choose column A_j to be any column with a negative reduced cost. Let $\vec{u} = B^{-1}A_j$ be the jth column of the tableau. We will have x_j enter the basis (this can be chosen using any rule).
2. For all i such that $u_i > 0$, divide the ith row of the tableau by $u_i$and choose the lexicographically smallest row. This row corresponds to the variable to leave the basis.

Example

Suppose we have the following tableau (we will do dots to avoid writing the entire row. The later columns will not be needed for the example).

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & \dots \\ & 0 & 1 & 1 & -1 & \dots \\ x_5 = & 1 & 0 & 5 & 3 & \dots \\ x_6 = & 2 & 4 & 6 & -1 & \dots \\ x_7 = & 3 & 0 & 7 & 9 & \dots \end{bmatrix}$$

Then we may select $j = 3$ to enter the basis as $\bar{c}_3 < 0$. In the corresponding column, we see that $u_1 = 3, u_2 = -1, u_3 = 9$. As $u_2 < 0$, we can ignore this row safely. We will divide the entire 1st and 3rd rows by u_1 and u_3 respectively to apply the lexicographic ordering rule.

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & \dots \\ & - & - & - & - & \dots \\ x_5 = & \frac{1}{3} & 0 & \frac{5}{3} & 1 & \dots \\ x_6 = & - & - & - & - & \dots \\ x_7 = & \frac{1}{3} & 0 & \frac{7}{9} & 1 & \dots \end{bmatrix}$$

Notice that row 3 is the smallest lexicographically of all the relevant rows. This means we select the variable in the third row to leave the basis (Here, we used a $-$ for the rows that we are not considering). As we have a BFS, we know A will be of rank m. Since A has m L.I. rows, this implies that we will never have a row that is a constant multiple of another. As such, we can always find a unique row that is smallest lexicographically!

Theorem

Suppose the simplex method in tableau form is used. Suppose all rows (except possibly the 0th row) in the simplex tableau are lexicographically positive. Also suppose we use the lexicographic pivoting rule. Then the following holds:

1. Every row (except possibly the 0th row) of the simplex tableau remains lexicographically positive after each iteration of the simplex algorithm.
2. The 0th row will strictly increase lexicographically after each iteration of the simplex method.
3. The simplex method will terminate after finitely many iterations (i.e. no cycling!)

Proof

1) Suppose all rows (except possibly the 0th row) in the simplex tableau are lexicographically positive. Suppose x_j is entering the basis and the lth row is the pivot row. Then we have $u_l > 0$ and if R_k is the kth row, we see

$$\frac{R_l}{u_l} \overset{L}{<} \frac{R_i}{u_i}$$

if $i \neq l$ and $u_i > 0$. If we wish to get the new tableau, we need to add some multiple of the lth row to the ith row in order to row reduce. We will denote the new kth row as R'_k. If $i = l$, then we see $\frac{1}{u_l} > 0$, so we note that

$$R_l \overset{L}{>} \vec{0}$$

implies

$$R_l' = \frac{1}{u_l} R_l \overset{L}{>} \frac{1}{u_l} \vec{0} = \vec{0}$$

so this row remains lexicographically positive. Now, consider the cases where $i \neq l$. If $u_i = 0$, then we have

$$R_i' = R_i - \frac{u_i}{u_l} R_l = R_i - 0 = R_i \overset{L}{>} \vec{0}$$

Now, suppose $u_i < 0$. Then we see

$$R_i' = R_i - \frac{u_i}{u_l} R_l \overset{L}{>} \vec{0}$$

as $-\frac{u_i}{u_l} > 0$ and thus $R_i, -\frac{u_i}{u_l} R_l \overset{L}{>} \vec{0}$. Now, suppose $u_i > 0$. Then

$$\frac{R_l}{u_l} \overset{L}{<} \frac{R_i}{u_i}$$

So

$$\frac{u_i}{u_l} R_l \overset{L}{<} R_i$$

$$\vec{0} \overset{L}{<} R_i - \frac{u_i}{u_l} R_l = R_i'$$

In all cases, we see $R_i' \overset{L}{>} \vec{0}$.

2) As we selected column, j, this implies that $\bar{c}_j < 0$. If we row reduce, this means the new 0th row is found by computing

$$R_0' = R_0 - \frac{\bar{c}_j}{u_l} R_l$$

but as $-\frac{\bar{c}_j}{u_l} > 0$, then $-\frac{\bar{c}_j}{u_l} R_l \overset{L}{>} \vec{0}$, thus

$$R_0' = R_0 - \frac{\bar{c}_j}{u_l} R_l \overset{L}{>} R_0 + \vec{0} = R_0$$

so the 0th row is strictly increasing lexicographically.

3) As there are only finitely many ways to select the basis of the tableau (including permutations of the variable orders) and as the tableau is determined by the selected basis, then there can only be finitely many 0th rows. As such, the simplex algorithm must terminate (as it is unable to cycle). Note that this theorem assumes a "nice" starting tableau. Fortunately there is a method to ensure that such a tableau is always chosen. We will talk more about this in a future lecture.

Bland's Rule

The other common method for cycle avoidance is **Bland's Rule**. It is stated as follows:

1. First the smallest j such that $\bar{c}_j < 0$ and have the corresponding column A_j enter the basis.
2. Out of all the variables x_i such that $\theta^* = \frac{x_{B(i)}}{u_i}$, select the smallest such i. Have $B(i)$ exit the basis.

I claim without proof that this method will not result in cycling. Typically we use the lexicographic pivoting rule for when we are working with the simplex tableau and we use Bland's rule when we are working with the revised simplex rule.